3 Fig. 3.1 shows a mains electric heater used to heat a small room.

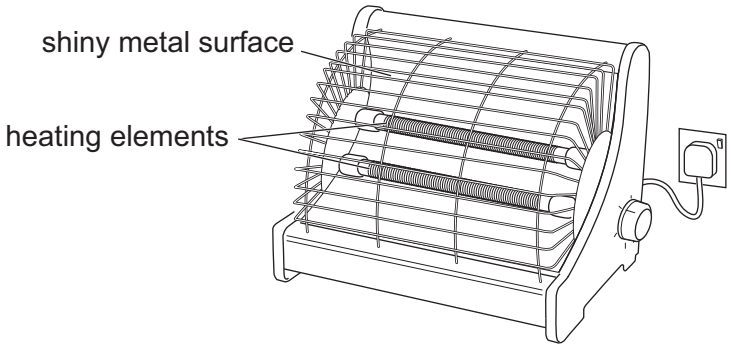

Fig. 3.1

(a) State the region of the electromagnetic spectrum which radiates thermal energy from the heater.

..... [1]

(b) Explain why the shiny metal surface behind the heating elements increases the thermal energy radiated into the room.

.....

.....

..... [2]

(c) The metal outer casing of the heater is earthed. State why this is an important safety feature.

.....

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(d) The mains voltage is 230 V. The two identical heating elements are connected in parallel. Each heating element has a resistance of 89 Ω.

(i) Calculate the current in **one** heating element.

current = ... [2]

(ii) Show that the electrical power of the heater is approximately 1200 W. State any equation you use in words or symbols.

[2]

(iii) The heater is 95% efficient at converting electrical work done to thermal energy.

Calculate the thermal energy emitted by the heater in **(d)(ii)** in 60 s. Give your answer to **two** significant figures.

thermal energy = ... [3]

[Total: 11]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN